Single-step Analysis with Guimera's Model

Zeyuan Chen

March 24, 2026

1 Two Models of Team Recruitment

1.1 Notation

There are two genders in the population: males constitute the majority, with a proportion of $p > 0.5$, while females account for $1 - p < 0.5$. For an individual, when selecting a partner, there exists a homophily coefficient for each gender $h_m, h_f \in [0, 1]$ that weights the probability of selecting a partner based on a preference for the same-gender (In the derivation, sometimes h instead of h_m, h_f is used, which assumes that $h_m = h_f$; the case where $h_m \neq h_f$ will also be discussed accordingly.). Specifically, the probability that a male selects another male as a partner is:

$$\pi_m = \frac{ph_m}{ph_m + (1-p)(1-h_m)}$$

The probability that a female selects another male as a partner is:

$$\pi_f = \frac{p(1-h_f)}{p(1-h_f) + (1-p)h_f}$$

1.2 Single-Seed Model

1.2.1 Model Setting

The recruiting process is:

- Initially, randomly sample an author with probabilities of $\{p, 1-p\}$ as the first team member.

- When recruiting the k-th member, the seed author will always be the first team member. If this seed author is male, the probabilities that he will use to select member k are $\{\pi_m, 1-\pi_m\}$; otherwise, the probabilities are $\{\pi_f, 1-\pi_f\}$.

1.2.2 General Term by T

Essentially, what we need to find is the expected number of male authors in the team after T rounds of recruitment, denoted as $E[M_T]$. To calculate it, we first calculate the expected number of males, $E[M_T^m]$ and $E[M_T^f]$, in male-seeded and female-seeded teams of size T, respectively. Clearly, we have:

$$\begin{cases} E[M_T^m] = 1 + (T-1)\pi_m \\ E[M_T^f] = (T-1)\pi_f \end{cases}$$

Therefore,

$$\begin{aligned} E[M_T] &= p \cdot E[M_T^m] + (1-p) \cdot E[M_T^f] \\ &= p + (T-1)[p\pi_m + (1-p)\pi_f] \end{aligned}$$

The expected ratio of males in a team with size T, denoted as r_T, is:

$$r_T = \frac{E[M_T]}{T} = \frac{p}{T} + \frac{T-1}{T}[p\pi_m + (1-p)\pi_f]$$

After substituting π_m, π_f, we finally arrive at the general formula:

$$r_T = p + \frac{T-1}{T} \cdot \frac{p(2p-1)}{1-p} \cdot \frac{(1-h)(2h-1)}{(hq+1-h)(q-hq+h)}$$

where $q = \frac{p}{1-p}$.

Therefore, the overrepresentation of males, i.e., $r_T - p > 0$, will always hold as long as $(2p-1)(2h-1) > 0$. In other words, as long as men constitute the majority of the population and there exists a homophily preference parameter $h > 0.5$, men will be overrepresented in sufficiently large groups.

r_T also increases monotonically with T; in other words, the larger the team, the stronger the overrepresentation of men within it. Specifically, this function converges to the limit at a rate of $O(1/T)$, where $\frac{T-1}{T} \to 0$. The limit r^* is:

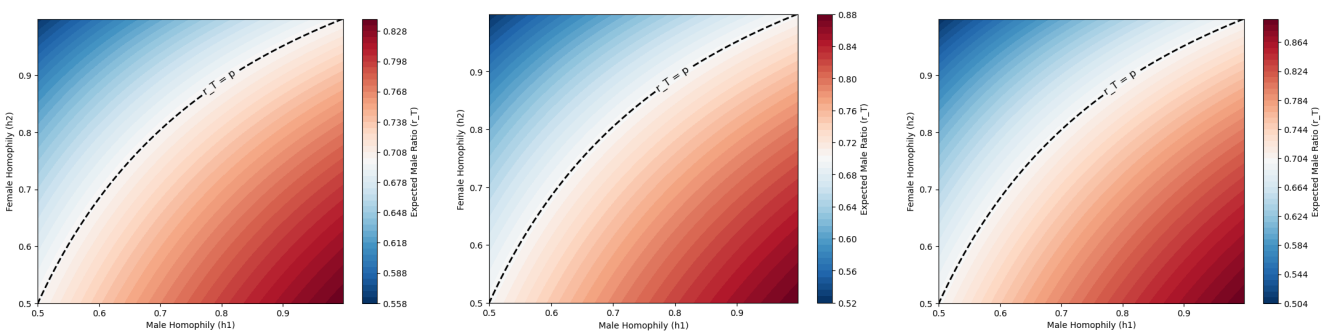
$$r^* = p\pi_m + (1-p)\pi_f$$

Therefore, the general term could also be expressed as:

$$r_T = \frac{p + (T-1)r^*}{T}$$

1.2.3 h_m and h_f

It is easy to prove that a sufficient and necessary condition of $r_T = p$ (equilibrium) is that $r^* = p$, which is irrelevant to team size T. Therefore, no matter how large the team size T is, the equilibrium conditions are always the same for h_m and h_f. However, team size T does indeed affect the state of imbalance. Specifically, larger teams lead to a greater overrepresentation of men in imbalanced settings, assuming p, h_m and h_f remain constant (Figure 1).



(a) T=3 (b) T=7 (c) T=15

Figure 1: How r_T Changes with h_m and h_f ($p = 0.7$) in Single-Seed Model

1.3 Team-Seed Model

1.3.1 Model Setting

The recruiting process is:

- Initially, randomly sample an author with probabilities of $\{p, 1-p\}$ as the first team member.
- When recruiting the $k+1$-th member, a member is randomly selected from the first k members to serve as the seed author. If this seed author is male, the probabilities that he will use to select member $k+1$ are $\{\pi_m, 1-\pi_m\}$; otherwise, the probabilities are $\{\pi_f, 1-\pi_f\}$.

1.3.2 Recurrence Relation

Essentially, what we need to find is the expected number of male authors in the team after T rounds of recruitment. Let $E[M_k]$ denote the expected number of males in the team after k rounds of recruitment (actually it is equal to assemble a team with size k); we then need to calculate $E[M_T]$.

Suppose that after the k-th recruitment, the team consists of M_k men and $k - M_k$ women. Therefore, the probability that the $(k+1)$-th member is a man is:

$$P(G_{k+1} = \text{male}|M_k) = \frac{M_k}{k}\pi_m + \frac{k - M_k}{k}\pi_f$$

Apparently, we have:

$$E[M_{k+1}] = E[M_k] + E[G_{k+1} = \text{male}]$$

Thus,

$$\begin{aligned} E[M_{k+1}] &= E[M_k] + E\left[\frac{M_k}{k}\pi_m + \frac{k - M_k}{k}\pi_f\right] \\ &= E[M_k]\left(1 + \frac{\pi_m - \pi_f}{k}\right) + \pi_f \end{aligned}$$

Specifically, when there is no homophily, i.e., $h = 0.5$, thus $\pi_m = \pi_f = p$, the recurrence formula reduces to:

$$\begin{aligned} E[M_{k+1}] &= E[M_k] + p \\ &= T \cdot p \end{aligned}$$

which is true.

1.3.3 Limit with Sufficiently Large T

As T approaches infinity, we can assume that the expected proportion of men in the team converges to a ratio $r^* \in [0, 1]$. In this case, the general formula can be rewritten as:

$$
\begin{aligned}
&E[M_{k+1}] = E[M_k]\left(1 + \frac{\pi_m - \pi_f}{k}\right) + \pi_f \\
&\Rightarrow (k+1)\cdot r^* = k \cdot r^* \left(1 + \frac{\pi_m - \pi_f}{k}\right) + \pi_f \\
&\Rightarrow r^* = \frac{\pi_f}{1 - \pi_m + \pi_f}
\end{aligned}
$$

It can be shown that r^* is monotonic with respect to h and p, and that a necessary and sufficient condition for $r^* > p$ is that $(2h-1)(2p-1) > 0$. In other words, as long as men constitute the majority of the population and there exists a homophily preference parameter $h > 0.5$, men will be overrepresented in sufficiently large groups.

When the homophily parameters for male and female are not identical, i.e. $h_m \neq h_f$, a necessary and sufficient condition for $r^* > p$ is:

$$
ph_m - (1-p)h_f > (2p-1)(2h_m h_f - h_m - h_f + 1)
$$

When $p = 0.5$, this inequality will hold as long as $h_m > h_f$, i.e., male exhibits a higher homophily than female does. When $p > 0.5$, even if h_m is slightly lower than h_f, male may still hold a overrepresented position in the team's gender composition.

1.3.4 General Term by T

Apart from extreme cases of sufficiently large teams, we are more interested in how the male representation advantage changes with team size, especially when T is not sufficiently large (e.g., $T < 10$, which is more common in the empirical data).

For simplicity, we denote $\alpha = \pi_m - \pi_f$, with α necessarily positive and smaller than 1, indicating the difference of the probabilities of selecting a male coauthor for men and women respectively. Therefore, the recurrence formula is transformed as:

$$
E[M_{k+1}] = E[M_k]\left(\frac{k+\alpha}{k}\right) + \pi_f
$$

And the limit r^* is transformed as:

$$
r^* = \frac{\pi_f}{1-\alpha}
$$

If we explode the recurrence relation to the T-th term, then:

$$E[M_T] = p \cdot \prod_{j=1}^{T-1} \left(\frac{j+\alpha}{j} \right) + \pi_f \cdot \sum_{i=1}^{T-1} \left(\prod_{j=i+1}^{T-1} \left(\frac{j+\alpha}{j} \right) \right)$$

Rewriting using the Gamma function $\Gamma(\cdot)$, the general term formula becomes:

$$\begin{aligned} E[M_T] &= p \cdot \frac{\Gamma(T+\alpha)}{\Gamma(1+\alpha)\Gamma(T)} + \pi_f \sum_{i=1}^{T-1} \frac{\Gamma(T+\alpha)\Gamma(i+1)}{\Gamma(i+1+\alpha)\Gamma(T)} \\ &= \frac{\Gamma(T+\alpha)}{\Gamma(T)} \left(\frac{p}{\Gamma(1+\alpha)} + \pi_f \sum_{i=1}^{T-1} \frac{\Gamma(i+1)}{\Gamma(i+1+\alpha)} \right) \end{aligned}$$

Considering this identity (using $\Gamma(y+1) = y\Gamma(y)$ for any y):

$$\frac{\Gamma(x)}{\Gamma(x+\alpha+1)} = \frac{1}{\alpha} \left(\frac{\Gamma(x)}{\Gamma(x+\alpha)} - \frac{\Gamma(x+1)}{\Gamma(x+1+\alpha)} \right)$$

We have:

$$\begin{aligned} E[M_T] &= \frac{\Gamma(T+\alpha)}{\Gamma(T)} \left(\frac{p}{\Gamma(1+\alpha)} + \frac{\pi_f}{1-\alpha} \left(T \frac{\Gamma(T)}{\Gamma(T+\alpha)} - \frac{1}{\Gamma(1+\alpha)} \right) \right) \\ &= \frac{\Gamma(T+\alpha)}{\Gamma(T)} \left(\frac{p}{\Gamma(1+\alpha)} + r^* \left(T \frac{\Gamma(T)}{\Gamma(T+\alpha)} - \frac{1}{\Gamma(1+\alpha)} \right) \right) \\ &= T \cdot r^* + (p - r^*) \frac{\Gamma(T+\alpha)}{\Gamma(T)\Gamma(1+\alpha)} \end{aligned}$$

where we denote that:

$$r_T = \frac{E[M_T]}{T} = r^* + (p - r^*) \frac{\Gamma(T+\alpha)}{\Gamma(1+T)\Gamma(1+\alpha)}$$

Recall that r^* is the limit when T is large, and that $p - r^*$ is constantly negative as $r^* > p$. Therefore, when T is not very large, the male representation (r_T) in the team is weaker than when T is sufficiently large (r^*). We are further concerned with a) whether the male representation is still higher than the overall male proportion p when T is not very large ($r^T > p$); and b) how fast r_T converges by T, i.e., at what rate the male representation advantage ($r_T - p$) increases with T.

For question a, it is equal to prove this inequality (with $T > 1, 0 < \alpha < 1$):

$$
\begin{aligned}
& r^* - p > (r^* - p)\frac{\Gamma(T+\alpha)}{\Gamma(1+T)\Gamma(1+\alpha)} \\
& \Leftarrow 1 > \frac{\Gamma(T+\alpha)}{\Gamma(1+T)\Gamma(1+\alpha)} \\
& \Leftarrow T \cdot \alpha \cdot \Gamma(T) \cdot \Gamma(\alpha) > \Gamma(T+\alpha)
\end{aligned}
$$

which is provable, as the equality only holds when $T = 1$ or $\alpha = 1$.

For question b, we can look only at the component related to T in the general term, that is:

$$
O\left(\frac{\Gamma(T+\alpha)}{\Gamma(1+T)}\right)
$$

Recall that in the Single-Seed Model, the component related to T is:

$$
O\left(\frac{1}{T}\right) = O\left(\frac{\Gamma(T)}{\Gamma(1+T)}\right)
$$

And it is easy to show that the growth rate of $\Gamma(T+\alpha)$ is larger than $\Gamma(T)$ when $T \geq 2$. Therefore, in Team-Seed Model, although the male representation advantage still grows as T gets larger, the rate at which it grows is slightly lower than that in the single-seed-author model that we adopted previously.

1.3.5 h_m and h_f

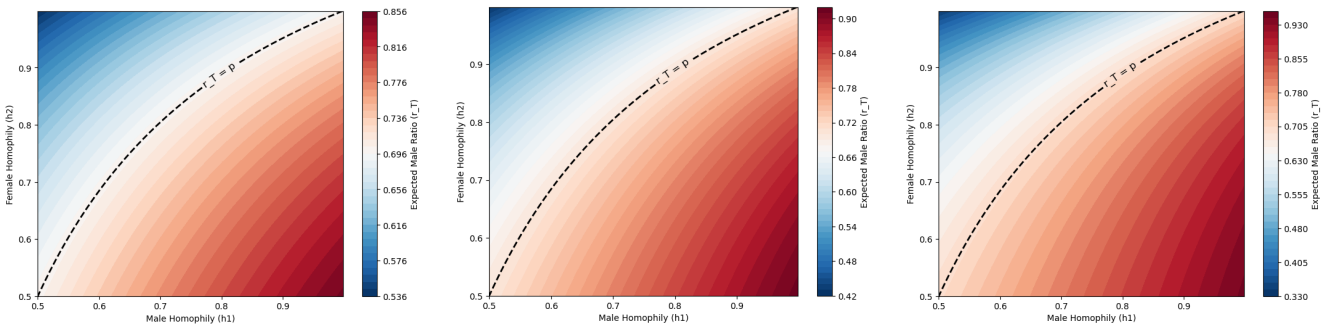
Recall that the state of equilibrium, i.e., $r_T = p$, only holds when (for any T):

$$
r^T - p = (r^* - p)\left[1 - \frac{\Gamma(T+\alpha)}{\Gamma(1+T)\Gamma(1+\alpha)}\right] = 0
$$

which is equal to:

$$
r^* = p
$$

Therefore, no matter how large the team size T is, the equilibrium conditions are always the same for h_m and h_f (as r^* is entirely determined by h_m and h_f). However, team size T does indeed affect the state of imbalance. Specifically, larger teams lead to a greater overrepresentation of men in imbalanced settings, assuming p, h_m and h_f remain constant (Figure 2).



(a) T=3 (b) T=7 (c) T=15

Figure 2: How r_T Changes with h_m and h_f ($p = 0.7$) in Team-Seed Model

2 Empirical Estimation

$$r_T = \omega r_T^{TS} + (1-\omega) r_T^{SS}$$

Three ways to estimate params:

- OLS:

$$\min_{h_m, h_f} \sum_{T=2}^{T_{max}} (\bar{r}_T^{observed} - r_T^{predicted}(h_m, h_f, p))^2$$

- MLE (but the likelihood function can be quite complex)
- Approximate Bayesian Computation